# PULSE_FACT: implemented model and candidate innovation sites

Source-grounded planning diagram | solid arrows: current data flow | orange labels: research opportunities

Inherited PULSE components

Adapted FACT interaction

Candidate only; not implemented

Cycle index
Historical timestamps

I1

**Historical phase reference**

Codebook + TimeEnc_x
A_x: [B,L,C]

A_x (not mixed by SAM)

I7

**Future reference / Phase Router**

A_y = Router(Y_0, A_x) + T_y
Patch-wise routing; actual attention heads = 1

**Known future**

TimeEnc_y
T_y: [B,H,C]

Candidate I4: phase-conditioned interaction

Current FACT block has no direct phase-conditioning input

Y_0

A_y

**History X**

[B,L,C]

I2

**Reference norm**

RevIN(X - A_x)
+ A_x

I3

**SAM**

Train only
Mix input + stats

**Temporal MLP**

Linear(L,512)
GELU

I4

**FACT block**

[B,C,512]
One block: N = 1

**Output map**

Linear(512,H)
Y_0: [B,H,C]

**Paired restoration**

RevIN inverse(Y_0 - A_y)
+ A_y

**Forecast**

Y_hat
[B,H,C]

Paired mean / variance (mixed in training)

**Inside the current FACT block (default settings)**

**Grid adapter**

Cyclic pad; reshape
[B,512,g,g]

**Parallel DW branches**

BN + kernels {1,3}
Dilations {1,2}; sum + U

**Pointwise FFN**

BN; 512 > 256 > 512
GELU; dropout 0.1; residual

**Crop + transpose**

g = floor(sqrt(C)) + 1
Back to [B,C,512]

I5 Candidate: stack / alternate blocks; N=2 or 3 is not implemented

I6 Candidate: selective residual or time / frequency branches (not the current model)

I8

**Training objective (outside Model.forward)**

lambda_rec * MSE + lambda_fft * mean |FFT(error)|
Current scripts: (lambda_rec, lambda_fft) = (0, 1)
Target uses the same SAM lambda and permutation

**Candidate losses / physical consistency**

Time + frequency balance; differences; phase-aware terms
Physical constraints require valid variables, units and laws

B=batch; L=history; H=horizon; C=variables. SAM bypassed at inference; A_x and timestamps remain unmixed. Grid is not a physical topology.